

INDIA@100 | ENERGY BATTERIES

ALL CHARGED UP

BATTERIES ARE CRITICAL TO INDIA'S PLANS TO CUT OIL BILLS AND GENERATE 500 GW OF GREEN ENERGY BY 2030

By **AJAY SUKUMARAN**

Fast facts first. About 140,000 electric vehicles were sold in India five years ago. In 2022-23, the number reached 1.17 million, thanks largely to a government scheme that incentivised production. E-vehicles are still a minuscule segment in a country where 19.6 million cars and two-wheelers were sold last year. But the goal is that by 2030, 30 per cent of all private cars and 80 per cent of two-wheelers sold will be electric.

Electric mobility, however, is only part of the big battery picture. In its transition to clean energy, India plans to generate 500 gigawatts (GW) of power from non-fossil fuel sources by 2030. Electricity generation from renewable sources such as solar and wind is always intermittent, unlike coal-based power. Hence, grid storage is going to be a crucial factor. And this is where batteries come in. Apart from running cars and consumer electronics, advanced lithium-ion (Li-ion)-based batteries can help make power grids more efficient by storing renewable energy for later use.

WHY IT IS A GAME CHANGER

Potent, secure and an exemplar of the circular economy, batteries might hold the key to decarbonising the domains of transportation and energy that span the whole planet.

NITI Aayog estimates that there will be a 50-100-fold increase from the existing domestic demand for batteries. By the end of this decade, the annual market for batteries in India—both for stationary grid usage and electric vehicles—will